

PROJECT DESIGN PHASE

Problem–Solution Fit

Project Title

SLA Breach Awareness and Justification System

Introduction

Organizations rely on Service Level Agreements (SLAs) to ensure timely resolution of incidents and service requests. However, teams often miss SLA deadlines due to a lack of timely alerts and monitoring. When SLA breaches occur, there is often no structured process to record valid reasons or obtain managerial approval. This reduces accountability and affects service quality.

The proposed ServiceNow solution addresses these challenges by providing SLA breach awareness through automated notifications, allowing users to submit breach justifications, enabling managers to review and approve or reject justifications, and generating reports for SLA performance monitoring.

Problems Identified

- Lack of awareness before SLA deadlines.
- Frequent SLA breaches due to delayed responses.
- No proper process to submit breach justifications.
- Lack of manager approval for SLA breach reasons.
- Poor SLA tracking and monitoring.
- Reduced accountability and transparency.
- Difficulty in generating SLA compliance reports.

Proposed Solution

The proposed ServiceNow application provides:

- Automated SLA monitoring.
- Early SLA breach notifications.
- SLA breach justification submission.
- Manager approval or rejection of justifications.
- Email notifications.
- SLA performance dashboards and reports.
- Improved accountability and compliance.

Problem–Solution Mapping

Problem	Proposed Solution
1.Lack of SLA awareness	Automated SLA alerts and notifications
2.Frequent SLA breaches	Early warning system
3.No justification process	SLA Breach Justification Form
4.No approval workflow	Manager Approval Workflow
5.Poor SLA monitoring	SLA dashboards and reports
6.Lack of accountability	Justification tracking and audit records

Expected Benefits

- Reduces SLA breaches.
- Improves SLA compliance.
- Faster response to approaching deadlines.
- Better accountability and transparency.
- Improved service quality.
- Simplified reporting and auditing.

Proposed Solution

Component 1: SLA Breach Detection Layer

- Business Rules monitor all active Task SLA records on Incident, Problem, and Change tables.
- Scheduled Jobs run at configurable intervals (every 5 minutes) to identify records where SLA_percentage_complete >= 75%.
- Breach events are created in a custom 'u_sla_breach_event' table with details: record number, assigned group, SLA name, breach time, and severity.

Component 2: Flow Designer Orchestration

- A master Flow Designer flow (Subflow: 'SLA Breach Orchestrator') is triggered by the breach event table insert.
- The flow determines the breach severity tier (Warning 75%, Critical 90%, Breached 100%) and routes to appropriate subflows.
- Parallel branches handle: (a) VA conversation initiation, (b) stakeholder email notifications, (c) escalation scheduling.

Component 3: Virtual Agent Conversation

- The VA Topic 'SLA Breach Justification' is programmatically triggered for the assigned agent/group using the VA API (sys_cs_conversation table insert).
- NLU model trained on breach-related intents: 'provide_justification', 'request_extension', 'escalate_breach', 'acknowledge_breach'.
- Conversation flow collects: Justification Category (dropdown), Detailed Reason (free text), Expected Resolution Time, and Action Plan.
- Collected data is structured into a JSON payload and written back to the Incident record via a Scripted REST API call.

Component 4: Stakeholder Notification & Escalation

- Email notifications sent to IT Manager with breach details, assigned agent, SLA name, and breach duration.
- Tiered escalation: unacknowledged breaches auto-escalate to Team Manager after 15 minutes and to IT Director after 30 minutes.
- Microsoft Teams integration (via IntegrationHub) posts breach alert cards to the team channel.
- Component 5: Reporting & Analytics Dashboard
- Performance Analytics indicators track: Total Breaches, Breach by Priority, Breach by Team, Breach by Category, Repeat Breach Rate, and Mean Time to Justify (MTTJ).
- Dashboards provide daily/weekly/monthly views accessible to IT Managers and Directors with drill-down capability.

BACKEND CONFIGURATION & DATA SETUP

1 Update Set Creation



Objective

To capture all customizations made in the ServiceNow instance so they can be migrated to another instance without losing changes.

Description

An Update Set is created before starting development. It records all configurations such as tables, fields, business rules, UI policies, Flow Designer flows, notifications, Virtual Agent topics, and other customizations related to the SLA Breach Awareness and Justification System.

Steps

1. Navigate to System Update Sets → Local Update Sets.
2. Click New.
3. Name: SLA Breach Awareness and Justification System.
4. Set the state to In Progress.
5. Make it the Current Update Set.
6. Perform all project configurations under this update set.

Expected Outcome

All project customizations are stored in a single update set and can be moved to another instance.

2 Create Business Schedule



Objective

To define the organization's working hours for SLA calculations.

Description

A business schedule is created so SLA timers calculate only during official working hours.

Business Schedule Configuration

Day	Working Hours	Type
Monday	09:00 AM – 06:00 PM	Working
Tuesday	09:00 AM – 06:00 PM	Working
Wednesday	09:00 AM – 06:00 PM	Working
Thursday	09:00 AM – 06:00 PM	Working
Friday	09:00 AM – 06:00 PM	Working
Saturday	–	Holiday
Sunday	–	Holiday

Expected Outcome

SLA time is calculated only during business hours.

3 SLA Creation



Objective

To create an SLA for Incident records.

Description

An SLA Definition is created to monitor the response and resolution time of incidents.

Example SLA Configuration

Field	Value
Name	Incident Resolution SLA
Table	Incident
Type	Resolution
Duration	8 Hours
Schedule	Business Schedule
SLA Timer	Pauses when conditions are met

Expected Outcome

The SLA starts automatically when an incident is created.

4 Configure SLA Start / Stop / Pause Conditions



Objective

To define when the SLA starts, pauses, and stops.

Configuration

Condition Type	When Condition Met	Example Configuration
Start Condition	When the record is created or criteria match	- Active = True - State = New
Pause Condition	When work is on hold	State = On Hold
Stop Condition	When the work is completed	- State = Resolved - State = Closed

Expected Outcome

The SLA behaves correctly based on the incident status.

5 Create Required Incident Fields – SLA Justification Fields



Objective

To capture the reason whenever an SLA breach occurs.

Description

Custom fields are created on the Incident table to capture SLA breach justification and approval details.

Fields to be Created

Field Name	Type	Description
SLA Breached	True/False	Indicates whether SLA is breached
Breach Justification	Multi-line Text	Reason for SLA breach
Justification Status	Choice	Pending, Approved, Rejected
Manager Comments	Multi-line Text	Comments from manager
Approved By	Reference (User)	Manager who approved/rejected
Approval Date	Date/Time	Date and time of approval/rejection

Expected Outcome

Users can submit SLA breach reasons, and managers can review and approve or reject them.

Conclusion

The proposed solution effectively addresses SLA management challenges by integrating automated alerts, justification workflows, approval mechanisms, and reporting capabilities within the ServiceNow platform, improving operational efficiency and service quality.